Degree bound for Lasserre relaxation under regular sequence assumptions without infinity condition

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Problem

Let

$$I := (g_1, \ldots, g_k) \subset \mathbb{R}[x_1, \ldots, x_n], \qquad S := \{x \in \mathbb{R}^n : \ g_1(x) = \cdots = g_k(x) = 0, \ h_1(x) \geq 0, \ldots, h_m(x) \geq 0\},$$

and let

$$p := f - f^*,$$

where $f^* = \min_{x \in S} f(x)$ is assumed finite and attained on the nonempty feasible set S.

The given hypotheses are:

- $k \geq n$,
- $V_{\mathbb{C}}(I)$ is finite and nonempty,
- the homogenizations $\bar{g}_1, \ldots, \bar{g}_n \in \mathbb{R}[x_0, \ldots, x_n]$ form a regular sequence,
- but we do *not* assume the stronger "no solutions at infinity" condition used in Hua–Qu.

The question is to give an explicit order d such that

$$p \in I_{2d} + QM(1, h_1, \ldots, h_m)_{2d},$$

under the same sort of nonsingularity assumption at points where $p = 0$ that the paper uses to guarantee exactness at some finite order.

I will prove an explicit degree bound that does *not* use the H-basis property at all. The bound is coarse (because the final step uses a general Hermann-type ideal-membership estimate), but it is completely explicit and rigorous.

Alternative readings considered

There is one genuine ambiguity: when $k > n$, what exactly is the "same type of nonsingularity/rank assumption"? The strongest natural reading is the scheme-theoretic one:

> every complex zero $\zeta \in V_{\mathbb{C}}(I)$ with $p(\zeta) = 0$ is a reduced point of $I_{\mathbb{C}}$.

This is the clean zero-dimensional analogue of the paper's requirement that all complex points attaining f^* be nonsingular. I adopt this strongest reading. It implies the algebraic condition $(p) + (I : p) = \mathbb{R}[x]$, and my main theorem is stated in that more flexible algebraic form.

Related literature found

- **Hua–Qu (2025)** — *PARTIAL.* They prove an explicit exactness bound under the *stronger* assumption that the first n equality constraints have no nontrivial common zero at infinity, and the published abstract states this explicitly. In the arXiv/HTML version they also explicitly note that "it would be interesting to investigate the degree bound under the regular sequence condition only." (pubsonline.informs.org)
- **Nie (2013)** — *PARTIAL.* This proves finite convergence for Lasserre's hierarchy when the real variety is finite, but does not give the sort of explicit exactness degree sought here. (epubs.siam.org)
- **Baldi–Krick–Mourrain (2025)** — *PARTIAL/TOOL.* Their abstract shows an effective Positivstellensatz for finite semialgebraic sets under the algebraic condition $(f)+(I:f)=(1)$, with degree bounds linear in a quotient-basis degree. This is very close in spirit to the local-Artinian step below, but I do *not* import their proof. (arxiv.org)
- **Aschenbrenner (2004)** — *TOOL.* The paper gives a Hermann-type bound for solving polynomial linear systems over a field; I use its Theorem 3.4 in the final ideal-membership step. (math.ucla.edu)
- **Atiyah–Macdonald** — *TOOL/BACKGROUND.* I use the structure theorem for Artin rings and the fact that in an Artin local ring the maximal ideal is nilpotent. (math.ens.psl.eu)
- **Hashemi–Javanbakht–Möller (2018)** — *BACKGROUND.* This paper discusses degree bounds for H-bases and is useful context for why losing the H-basis property is a real obstruction. My proof avoids H-bases entirely. (lematematiche.dmi.unict.it)

Proof sketch

The proof has two independent parts.

First, I work *modulo the full equality ideal I*, inside the finite-dimensional real algebra

$$A := \mathbb{R}[x]/I.$$

If p satisfies the zero-dimensional nonsingularity condition $(p)+(I:p)=\mathbb{R}[x]$, then in every Artin-local factor of A, the image of p is either 0 or a unit. This allows a purely local construction: on a feasible real factor, a positive unit is a square; on an infeasible real factor, a negative unit becomes a square after dividing by one negative inequality h_j; on a complex factor, any unit is a square. Gluing these local squares with orthogonal idempotents yields

$$p \equiv q_0^2 + q_1^2 h_1 + \cdots + q_m^2 h_m \pmod{I}$$

in the quotient ring.

Second, the above identity is only modulo I. So I must rewrite the remainder in the *original generators* $g_1, \ldots, g_k$ with an explicit degree bound. Here I deliberately use brute force: a Hermann-type theorem gives an explicit bound for solving

$$\lambda_1 g_1 + \cdots + \lambda_k g_k = r.$$

This is exactly the place where Hua–Qu used the H-basis property; my replacement is much coarser, but it is fully effective and needs no assumption at infinity.

Finally, I need an explicit bound on the degrees of representatives in A. Under the regular-sequence hypothesis on $\bar{g}_1, \dots, \bar{g}_n$, a Hilbert-series argument gives

$$\dim_{\mathbb{R}} \big(\mathbb{R}[x]/(g_1, \dots, g_n)\big) \le \prod_{i=1}^{n} \deg g_i,$$

hence also

$$\dim_{\mathbb{R}}(A) \le \prod_{i=1}^{n} \deg g_i.$$

A Cayley–Hamilton reduction then shows that every class in A has a representative of degree at most

$$n\Big(\prod_{i=1}^{n} \deg g_i - 1\Big).$$

That is enough to close the argument.

Main result

Theorem 1. *Let*

$$I = (g_1, \dots, g_k) \subset \mathbb{R}[x_1, \dots, x_n], \qquad p := f - f^*,$$

and define

$$D_g := \max_{1 \le i \le k} \deg g_i, \qquad D_h := \max_{0 \le j \le m} \deg h_j \quad (h_0 := 1).$$

Assume $V_{\mathbb{C}}(I)$ is finite and that

$$(p) + (I : p) = \mathbb{R}[x].$$

Let $\beta \ge 0$ be any integer such that every class in $A := \mathbb{R}[x]/I$ has a representative of degree at most β. Set

$$\Delta(\beta) := \max\{\deg f,\ 2\beta + D_h,\ D_g\}, \qquad T(\beta) := D_g + \big(2\Delta(\beta)\big)^{2^n}.$$

Then

$$p \in I_{T(\beta)} + QM(1, h_1, \dots, h_m)_{T(\beta)}.$$

Hence Lasserre's relaxation is exact and attained for every order

$$d \ge \left\lceil \frac{T(\beta)}{2} \right\rceil.$$

Corollary 1. *Assume in addition that every $\zeta \in V_{\mathbb{C}}(I)$ with $p(\zeta) = 0$ is a reduced point of $I_{\mathbb{C}}$, and that $\bar{g}_1, \dots, \bar{g}_n$ form a regular sequence. Write*

$$\Pi := \prod_{i=1}^{n} \deg g_i, \qquad \beta_0 := n(\Pi - 1).$$

Then

$$p \in I_{T(\beta_0)} + QM(1, h_1, \dots, h_m)_{T(\beta_0)},$$

so exactness and attainment hold at every order

$$d \ge \left\lceil \frac{T(\beta_0)}{2} \right\rceil.$$

Equivalently, with

$$\Delta_0 := \max\{\deg f,\ 2n(\Pi - 1) + D_h,\ D_g\}, \qquad T_0 := D_g + \big(2\Delta_0\big)^{2^n},$$

one has

$$f - f^* \in I_{T_0} + QM(1, h_1, \ldots, h_m)_{T_0},$$

hence order $d_0 := \lceil T_0/2 \rceil$ is exact and attained.

Remark. If one also invokes the classical Gröbner-basis fact that a zero-dimensional quotient has a standard monomial basis, then the Cayley–Hamilton estimate $\beta_0 = n(\Pi - 1)$ can be sharpened to $\beta_0 = \Pi - 1$. This only improves constants; it is not needed for the proof below. A standard-bases note stating the relevant basis fact is available online. (homepages.math.uic.edu)

Proof

Lemma 1. *Let A be a finite-dimensional commutative $\mathbb{R}$-algebra, and let $a \in A$.*

1. *A is a finite direct product of Artin local rings:*

$$A \cong A_1 \times \cdots \times A_s.$$

2. *The following are equivalent:*

$$(a) + (0 :_A a) = A$$

and

$$\text{for every } \nu, \text{ the component } a_\nu \in A_\nu \text{ is either } 0 \text{ or a unit.}$$

Proof. Since A is finite-dimensional as an $\mathbb{R}$-vector space, every descending chain of ideals is a descending chain of subspaces, hence stabilizes. So A is Artinian. By the structure theorem for Artin rings, $A \cong \prod_{\nu=1}^{s} A_\nu$ with each A_ν Artin local; moreover the maximal ideal of each A_ν is nilpotent. Hypothesis audit: the cited theorem requires only that the ring be Artinian, and that is satisfied here because A is finite-dimensional over $\mathbb{R}$. (math.ens.psl.eu)

For (2), since ideals and annihilators in a finite product are computed componentwise, it is enough to treat the local case $A = A_\nu$. If $a = 0$, then $(a) + (0 : a) = A$. If a is a unit, then $(a) = A$. Conversely, suppose $a \neq 0$ is not a unit. In a local ring this means $a \in \mathfrak{m}$, where $\mathfrak{m}$ is the maximal ideal. If $b \in (0 : a)$ were a unit, then $ba = 0$ would force $a = 0$, contradiction. Hence $(0 : a) \subseteq \mathfrak{m}$, and also $(a) \subseteq \mathfrak{m}$. Therefore

$$(a) + (0 : a) \subseteq \mathfrak{m} \neq A.$$

So the equivalence holds. □

Lemma 2. *Let B be an Artin local $\mathbb{R}$-algebra with maximal ideal $\mathfrak{m}$.*

1. *If the residue field $B/\mathfrak{m} \cong \mathbb{R}$, then every unit whose residue is > 0 is a square in B.*

2. *If the residue field $B/\mathfrak{m} \cong \mathbb{C}$, then every unit is a square in B.*

Proof. Because B is Artin local, $\mathfrak{m}$ is nilpotent. Let $u \in B^\times$.

For (1), let $c \in \mathbb{R}_{>0}$ be the residue of u. Then

$$u = c(1+\eta)$$

for some nilpotent $\eta \in \mathfrak{m}$. Let N satisfy $\eta^{N+1} = 0$, and define

$$s(\eta) := \sum_{r=0}^{N} \binom{1/2}{r} \eta^r.$$

Since the formal binomial identity $\big((1+T)^{1/2}\big)^2 = 1+T$ truncates on nilpotents, we get $s(\eta)^2 = 1+\eta$. Hence

$$u = \big(\sqrt{c}\, s(\eta)\big)^2.$$

For (2), let $\bar{u} \in \mathbb{C}^\times$ be the residue of u. Choose $t \in \mathbb{C}$ with $t^2 = \bar{u}$, and choose any lift $v \in B$ of t. Then v is a unit and

$$uv^{-2} \equiv 1 \pmod{\mathfrak{m}},$$

so

$$uv^{-2} = 1 + \eta$$

with $\eta \in \mathfrak{m}$ nilpotent. Using the same truncated binomial series,

$$uv^{-2} = s(\eta)^2.$$

Therefore

$$u = (v\, s(\eta))^2.$$

$\square$

Lemma 3. *Let $A = \mathbb{R}[x]/I$, let $\bar{p}, \bar{h}_j$ denote the residue classes of p, h_j in A, and assume $(p) + (I : p) = \mathbb{R}[x]$. Let $\beta \geq 0$ be such that every class in A has a representative of degree $\leq \beta$. Then there exist polynomials $q_0, \ldots, q_m \in \mathbb{R}[x]$ with*

$$\deg q_j \leq \beta \qquad (0 \leq j \leq m)$$

such that

$$p - \Big(q_0^2 + \sum_{j=1}^{m} q_j^2 h_j\Big) \in I.$$

In particular,

$$p \in I_U + QM(1, h_1, \ldots, h_m)_U \quad \textit{for } U := \max\{\deg f,\ 2\beta + D_h\}.$$

Proof. Write

$$A \cong A_1 \times \cdots \times A_s$$

as in the previous lemma, and let $e_\nu \in A$ be the corresponding orthogonal idempotents. By Lemma 1, the hypothesis $(p) + (I : p) = \mathbb{R}[x]$ is equivalent to saying that in each local factor A_ν, the component $\bar{p}_\nu$ is either 0 or a unit.

Fix ν. There are three cases.

Case 1: the residue field of A_ν is $\mathbb{C}$. If $\bar{p}_\nu = 0$, set $u_\nu := 0$ and assign this factor to index 0. If $\bar{p}_\nu$ is a unit, then by Lemma 2(2) there exists $u_\nu \in A_\nu$ with

$$\bar{p}_\nu = u_\nu^2.$$

Again assign this factor to index 0.

Case 2: the residue field of A_ν is $\mathbb{R}$, corresponding to a feasible real point $x^{(\nu)} \in S$. Then $p(x^{(\nu)}) \geq 0$ by optimality of f^*. If $\bar{p}_\nu = 0$, set $u_\nu := 0$ and assign index 0. If $\bar{p}_\nu$ is a unit, then its residue is > 0, so by Lemma 2(1) there exists $u_\nu \in A_\nu$ with

$$\bar{p}_\nu = u_\nu^2.$$

Assign index 0.

Case 3: the residue field of A_ν is $\mathbb{R}$, corresponding to an infeasible real point $x^{(\nu)} \notin S$. Then some inequality is violated; fix one index $j(\nu) \in \{1, \ldots, m\}$ such that

$$h_{j(\nu)}(x^{(\nu)}) < 0.$$

Hence $\bar{h}_{j(\nu),\nu}$ is a unit of A_ν with negative residue. If $\bar{p}_\nu = 0$, set $u_\nu := 0$ and assign index 0. If $\bar{p}_\nu$ is a unit with positive residue, Lemma 2(1) gives $\bar{p}_\nu = u_\nu^2$, and we again assign index 0. If $\bar{p}_\nu$ is a unit with negative residue, then

$$\bar{p}_\nu \bar{h}_{j(\nu),\nu}^{-1}$$

has positive residue, so by Lemma 2(1) there exists $u_\nu \in A_\nu$ such that

$$\bar{p}_\nu = u_\nu^2 \, \bar{h}_{j(\nu),\nu}.$$

Assign this factor to index $j(\nu)$.

Now define, for $j = 0, \ldots, m$,

$$v_j := \sum_{\nu:\ \text{assigned to } j} e_\nu u_\nu \in A.$$

Because the e_ν's are orthogonal idempotents,

$$v_j^2 = \sum_{\nu:\ \text{assigned to } j} e_\nu u_\nu^2.$$

By construction, in A we therefore have

$$\bar{p} = v_0^2 + \sum_{j=1}^{m} v_j^2 \bar{h}_j.$$

By the definition of β, for each j there exists $q_j \in \mathbb{R}[x]$ with $\deg q_j \leq \beta$ whose residue class is v_j. Then

$$p - \Big(q_0^2 + \sum_{j=1}^{m} q_j^2 h_j\Big) \in I.$$

Also,

$$\deg(q_j^2 h_j) \leq 2\beta + \deg h_j \leq 2\beta + D_h,$$

and $\deg p \leq \deg f$. So the remainder lies in I and has degree at most

$$U := \max\{\deg f,\ 2\beta + D_h\}.$$

This proves the claim. □

Proof of Theorem. Apply the previous lemma to obtain $q_0, \dots, q_m$ with $\deg q_j \le \beta$ and

$$r := p - \Big(q_0^2 + \sum_{j=1}^m q_j^2 h_j\Big) \in I, \qquad \deg r \le U := \max\{\deg f,\ 2\beta + D_h\}.$$

Set

$$\Delta(\beta) = \max\{U, D_g\} = \max\{\deg f,\ 2\beta + D_h,\ D_g\}.$$

Now solve

$$\lambda_1 g_1 + \cdots + \lambda_k g_k = r.$$

This is a polynomial linear system with one equation and k unknowns over the field $\mathbb{R}$. By Aschenbrenner's version of Hermann's theorem, there exists a solution with

$$\deg \lambda_i \le (2\Delta(\beta))^{2^n} \qquad (1 \le i \le k).$$

Hypothesis audit: the theorem applies with base field $F = \mathbb{R}$, number of variables $N = n$, number of equations $m_{\text{lin}} = 1$, coefficient matrix $[g_1 \ \cdots \ g_k]$, and right-hand side r; every entry has degree $\le \Delta(\beta)$; and the system is solvable because $r \in I$. (math.ucla.edu)

Therefore

$$\deg(\lambda_i g_i) \le D_g + (2\Delta(\beta))^{2^n} = T(\beta).$$

Also,

$$\deg(q_0^2) \le 2\beta \le T(\beta), \qquad \deg(q_j^2 h_j) \le 2\beta + D_h \le T(\beta).$$

Hence

$$p = \sum_{i=1}^k \lambda_i g_i + q_0^2 + \sum_{j=1}^m q_j^2 h_j \in I_{T(\beta)} + QM(1, h_1, \dots, h_m)_{T(\beta)}.$$

By definition, this means Lasserre's SOS relaxation is exact and attained at every order d with $2d \ge T(\beta)$, i.e.

$$d \ge \left\lceil \frac{T(\beta)}{2} \right\rceil. \qquad \square$$

Lemma 4. *Assume $\bar g_1, \dots, \bar g_n$ form a regular sequence, and write*

$$d_i := \deg g_i, \qquad \Pi := \prod_{i=1}^n d_i, \qquad \delta := \sum_{i=1}^n (d_i - 1).$$

Then

$$\dim_{\mathbb{R}} \big(\mathbb{R}[x]/(g_1, \dots, g_n)\big) \le \Pi.$$

Consequently,

$$\dim_{\mathbb{R}}(\mathbb{R}[x]/I) \le \Pi.$$

Proof. Let $S := \mathbb{R}[x_0, \dots, x_n]$ and $\bar J := (\bar g_1, \dots, \bar g_n) \subset S$.

Because $\bar g_1, \dots, \bar g_n$ is a homogeneous regular sequence, for each $s = 1, \dots, n$ there is a short exact sequence

$$0 \to S/(\bar g_1, \dots, \bar g_{s-1})(-d_s) \xrightarrow{\cdot \bar g_s} S/(\bar g_1, \dots, \bar g_{s-1}) \to S/(\bar g_1, \dots, \bar g_s) \to 0.$$

Taking Hilbert series and iterating gives

$$HS_{S/\bar{J}}(z) = \frac{\prod_{i=1}^{n}(1-z^{d_i})}{(1-z)^{n+1}} = \frac{\prod_{i=1}^{n}(1+z+\cdots+z^{d_i-1})}{1-z}.$$

Let

$$Q(z) := \prod_{i=1}^{n}(1+z+\cdots+z^{d_i-1}) = \sum_{r=0}^{\delta} a_r z^r.$$

Then

$$HS_{S/\bar{J}}(z) = Q(z)\sum_{t\geq 0} z^t,$$

so the degree-t Hilbert function is

$$H(t) = \sum_{r=0}^{\min\{t,\delta\}} a_r.$$

Hence for every $t \geq \delta$,

$$H(t) = Q(1) = \prod_{i=1}^{n} d_i = \Pi.$$

Now let $J := (g_1, \ldots, g_n) \subset \mathbb{R}[x_1, \ldots, x_n]$, and for each $t \geq 0$ let

$$R_{\leq t} := \{u \in \mathbb{R}[x_1, \ldots, x_n] : \deg u \leq t\}.$$

Define

$$\Phi_t : \; R_{\leq t}/(J \cap R_{\leq t}) \longrightarrow (S/\bar{J})_t$$

by

$$u \longmapsto [x_0^{t-\deg u}\bar{u}].$$

This is well-defined: if $u \in J \cap R_{\leq t}$, say $u = \sum_{i=1}^{n} \lambda_i g_i$, then

$$x_0^{t-\deg u}\bar{u} = \sum_{i=1}^{n} x_0^{t-\deg(\lambda_i g_i)}\bar{\lambda}_i\,\bar{g}_i \in \bar{J}_t.$$

It is injective: if $x_0^{t-\deg u}\bar{u} \in \bar{J}_t$, then dehomogenizing at $x_0 = 1$ yields $u \in J$.

Therefore

$$\dim_{\mathbb{R}}\left(R_{\leq t}/(J \cap R_{\leq t})\right) \leq \dim_{\mathbb{R}}(S/\bar{J})_t \leq \Pi \qquad (t \geq \delta).$$

As t increases, the left-hand side is an increasing sequence of integers whose union is $\mathbb{R}[x]/J$. Hence it stabilizes, and the stabilized value is $\dim_{\mathbb{R}}(\mathbb{R}[x]/J)$. Thus

$$\dim_{\mathbb{R}}(\mathbb{R}[x]/J) \leq \Pi.$$

Since $I \supseteq J$, the quotient $\mathbb{R}[x]/I$ is a quotient of $\mathbb{R}[x]/J$, so

$$\dim_{\mathbb{R}}(\mathbb{R}[x]/I) \leq \Pi.$$

$\square$

Lemma 5. *Let $A = \mathbb{R}[x]/I$, and let $N := \dim_{\mathbb{R}} A$. Then every class in A has a representative of degree at most $n(N-1)$.*

Proof. For each $i = 1, \dots, n$, let

$$M_i : A \to A, \qquad M_i([u]) = [x_i u],$$

be multiplication by x_i. Since A is an N-dimensional vector space, the Cayley–Hamilton theorem gives a polynomial identity

$$M_i^N + c_{i,N-1} M_i^{N-1} + \cdots + c_{i,0} I_A = 0$$

for suitable real scalars $c_{i,r}$. Applying this identity to the class of $1 \in A$ gives

$$[x_i^N] = -\sum_{r=0}^{N-1} c_{i,r} [x_i^r].$$

So modulo I, every occurrence of x_i^N can be replaced by a linear combination of lower powers of x_i.

Now take any monomial $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$. If some $\alpha_i \geq N$, write

$$x^\alpha = x_i^N \, x_1^{\alpha_1} \cdots x_i^{\alpha_i - N} \cdots x_n^{\alpha_n},$$

and replace x_i^N using the above identity. Every resulting term has strictly smaller total degree, because x_i^N is replaced by a combination of x_i^r with $r \leq N-1$. Repeating finitely many times, we express the class of x^α as a linear combination of monomials with

$$0 \leq \alpha_i \leq N - 1 \qquad (1 \leq i \leq n).$$

Hence every class in A has a representative whose total degree is at most

$$n(N-1).$$

□

Lemma 6. *If every $\zeta \in V_{\mathbb{C}}(I)$ with $p(\zeta) = 0$ is reduced for $I_{\mathbb{C}}$, then*

$$(p) + (I : p) = \mathbb{R}[x].$$

Proof. Let $A = \mathbb{R}[x]/I$, and decompose it into local factors A_ν as above. By Lemma 1 it suffices to show that in each A_ν, the component of $\bar{p}$ is 0 or a unit.

If the corresponding point ζ satisfies $p(\zeta) \neq 0$, then the residue of $\bar{p}_\nu$ is nonzero, hence $\bar{p}_\nu$ is a unit.

If $p(\zeta) = 0$, the assumption says precisely that the local algebra of $I_{\mathbb{C}}$ at ζ is reduced, i.e. a field. Therefore the corresponding local factor has no nonzero nilpotents, so $\bar{p}_\nu$, having residue 0, must be 0.

Thus every component is 0 or a unit, and Lemma 1 yields $(p) + (I : p) = \mathbb{R}[x]$. □

Proof of the Corollary. By the last lemma, the reduced-point assumption implies $(p) + (I : p) = \mathbb{R}[x]$, so the theorem applies once we produce an explicit β.

By the previous lemma,

$$N := \dim_{\mathbb{R}}(\mathbb{R}[x]/I) \leq \Pi.$$

Hence every class in $\mathbb{R}[x]/I$ has a representative of degree at most

$$n(N-1) \leq n(\Pi - 1) = \beta_0.$$

Now apply the theorem with $\beta = \beta_0$. This gives

$$p \in I_{T(\beta_0)} + QM(1, h_1, \dots, h_m)_{T(\beta_0)},$$

which is exactly the claimed bound. □

Gap to the full statement

No logical gap remains for the *existence* of an explicit bound: the corollary gives one under the regular-sequence hypothesis and the same kind of nonsingularity assumption used in the paper.

What remains open is sharpness. The doubly exponential term enters only because I chose the most robust possible replacement for the missing H-basis property, namely Hermann's general ideal-membership bound. A more delicate analysis of ideal membership inside the complete-intersection-plus-extra-equations structure could substantially improve the constant, but this is a sharpening problem, not an existence problem.

Self red-team

Counterexample attempts.

- *Singular minimizer.* Take $I = (x^2) \subset \mathbb{R}[x]$, $m = 0$, $f = x$. Then $f^* = 0$ on the feasible set $\{0\}$, but $x \notin (x^2) + \Sigma[x]$. Here the point where $p = f - f^* = x$ vanishes is nonreduced, and indeed $(p) + (I : p) = (x) \neq (1)$. So the nonsingularity hypothesis is genuinely needed.
- *Points at infinity / failure of H-basis.* A typical example is

$$g_1 = x^2 - y, \qquad g_2 = xy - 1,$$

  whose highest-degree parts x^2 and xy have a nontrivial common zero, so there is a projective point at infinity. My proof never uses the H-basis property, so this obstruction does not affect it.
- *Boundary case $m = 0$.* Then the weighted local construction collapses to a plain square modulo I, and the argument still goes through unchanged.

Three likely referee objections.

1. *"Complex residue-field factors are not ordered; why is a unit a square there?"* I handled this explicitly by lifting a square root of the residue to the local ring and then taking a binomial square root of a 1+nilpotent element.
2. *"Regular sequence of $\bar{g}_1, \ldots, \bar{g}_n$ controls the homogeneous complete intersection, not obviously the affine quotient."* This is exactly why I inserted the injective homogenization map

$$R_{\leq t}/(J \cap R_{\leq t}) \hookrightarrow (S/\bar{J})_t.$$

   It transfers the complete-intersection Hilbert-function bound to the affine quotient without any H-basis assumption.
3. *"The extra generators $g_{n+1}, \ldots, g_k$ disappear in the quotient-dimension estimate."* They only help there, since $I \supseteq (g_1, \ldots, g_n)$ makes the quotient smaller. In the final certificate they are fully reinstated by the Hermann step, which rewrites the remainder in *all* original generators.

Degenerate cases checked.

- If $\Pi = 1$, then the first n equations are effectively linear and the quotient dimension bound gives $N \leq 1$, so $\beta_0 = 0$, as it should.

- If I is radical, the local argument simplifies: every local factor is a field, so the only obstruction is sign at real points.
- The proof does not accidentally imply exactness for the known bad example $I = (x^2), p = x$; the crucial algebraic hypothesis fails there.

Invoked results

1. **Structure theorem for Artin rings; nilpotence in the local case.** *Statement used.* Every Artin ring is a finite direct product of Artin local rings. In an Artin local ring, the maximal ideal is nilpotent. *Classification.* TOOL. *Citation.* Atiyah–Macdonald, *Introduction to Commutative Algebra*, Theorem 8.7 and the paragraph immediately preceding it. (math.ens.psl.eu) *Hypothesis audit.* I apply this only to $A = \mathbb{R}[x]/I$, which is finite-dimensional over $\mathbb{R}$ because $V_{\mathbb{C}}(I)$ is finite; therefore A is Artinian, so the theorem applies.
2. **Hermann–Seidenberg degree bound in Aschenbrenner's form.** *Statement used.* Let $D = F[X_1, \ldots, X_N]$ be a polynomial ring over a field F. If $A \in D^{m\times n}$ and $b \in D^m$ have degree at most d, and the system $Ay = b$ has a solution in D^n, then it has one of degree at most $(2md)^{2^N}$. *Classification.* TOOL. *Citation.* Aschenbrenner, Theorem 3.4. (math.ucla.edu) *Hypothesis audit.* I use it with $F = \mathbb{R}$, $N = n$, $m = 1$, coefficient matrix $[g_1 \ \cdots \ g_k]$, and right-hand side $r \in I$. All coefficients have degree at most $\Delta(\beta)$, and solvability is automatic because $r \in I$.
3. **(Optional sharpening only) Standard monomials of a zero-dimensional Gröbner basis form a basis of the quotient.** *Statement used.* For a zero-dimensional ideal I, monomials not contained in the leading-term ideal define a basis of the quotient ring. *Classification.* BACKGROUND/TOOL. *Citation.* Verschelde, course notes, p. 5. (homepages.math.uic.edu) *Hypothesis audit.* This is used only in the sharpening remark, not in the main proof.

Self-confidence

3 — I checked each step carefully and the proof is structurally robust, but the argument is long enough that a hidden bookkeeping mistake is still possible.